

HYBRID MODEL · LATE FUSION + SE GATING · SIDANG STUDY GUIDE

Built to fuse. Measured to **lose honestly.**

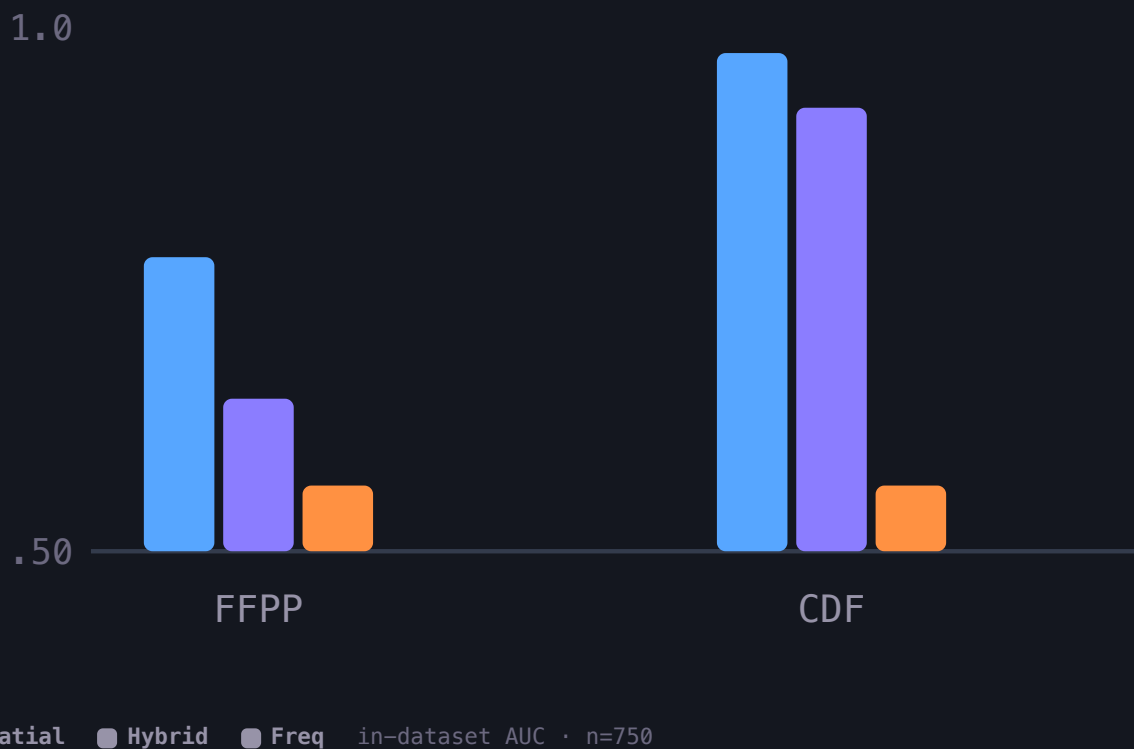
You own the **hybrid model** — `HybridTwoBranch`, the fusion design (projection → concatenation → SE gating → classifier), and the thesis's **headline comparative claim**: the proposed hybrid does **not** beat the spatial baseline. Your job at the defense is to defend that as a **rigorous ablation result**, not a failed improvement.

§0 · THE ONE-LINER YOU DEFEND

The hybrid is our **proposed** model. Its architecture is sound — we balance the two branches to 256-d each and add an **SE gate** to adaptively weight them. But because the frequency branch is near-random, fusion **injects noise**: the hybrid lands **below pure spatial** in-dataset on both datasets. The one place FFT helps is **reducing the generalization drop FFPP→CDF** — but only that direction, and at the cost of in-dataset accuracy. Our answer to “does adding FFT help?” is a **quantified, qualified no**.

Spatial > **Hybrid** > **Freq**

RANKING ON BOTH DATASETS, EVERY RELIABLE TIER



§1 What to study — the ordered curriculum

Each item: *what* · *why you need it* · *what to read*. The fusion is where you go deep — everything else you defer to the spatial and frequency guides.

1

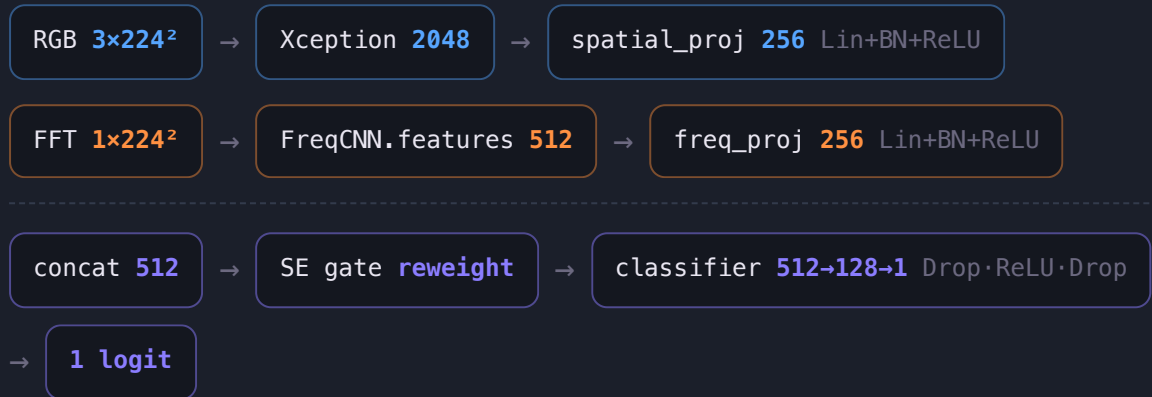
The full hybrid forward pass

HybridTwoBranch.forward · hybrid_fusion.py:56-67

1.5 HR

CORE

You must be able to draw this from memory, with dimensions.



- Two **independent** branches extract features (`self.spatial(rgb)` , `self.freq.features(fft)`), then fuse **once at the feature level** — that's **late fusion**.
- Spatial branch = Xception's **2048-d** global-avg-pooled feature (num_classes=0 extractor); freq branch = FreqCNN's **512-d** `.features` output (its classifier head bypassed).
- **~25.9M params** (25,878,570) = spatial extractor 20.8M + FreqCNN 4.2M + projections/SE/head (~0.85M).
- Thesis refs: **§3.4.3, Gambar 3.10** (architecture diagram).

2

Why project both branches to 256 — the symmetric bottleneck

45 MIN

CORE

This is the “why not just concatenate?” question.

- Raw dims are **wildly asymmetric: spatial 2048 vs freq 512 (a 4:1 ratio)**. Concatenating raw would let the 2048-d spatial branch **dominate**, and the gradient reaching the freq branch would shrink to near-nothing.
- `spatial_proj` / `freq_proj` each = `Linear(→256) + BatchNorm1d + ReLU`. Projecting **both to 256** equalizes their contribution; **BatchNorm** normalizes their scales; the result is a **balanced 512-d vector** the SE gate can weight fairly.
- **This is exactly the fix** for the original collapse (see item 7): the pre-fix model concatenated 2048:64 directly and spatial swamped everything (~97% spatial). Symmetric projection is *what makes late fusion fair between modalities*. (Thesis: projection eq. 2.15, §2.8.)

3

The SE gate — squeeze, excitation, scale

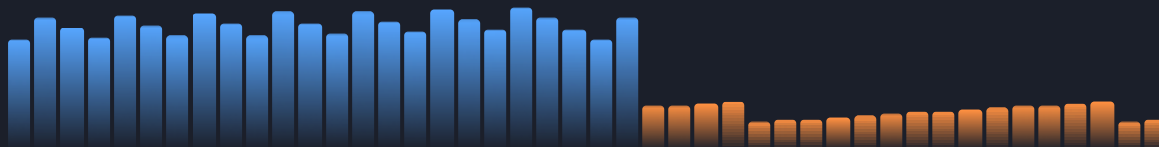
SEGate · hybrid_fusion.py:11-24

1 HR

CORE

The most likely “explain your fusion mechanism” question. SE-Net is Hu et al. [38].

- It's **channel attention** on the **512-d fused vector** (here each “channel” is a fused feature dimension).
- **Squeeze:** `Linear(512→128)` (reduction=4) summarizes cross-channel interactions into a compact descriptor.
- **Excitation:** `Linear(128→512) → Sigmoid` produces a per-dimension **gate weight in (0,1)**.
- **Scale (reweight):** multiply the fused vector by those gates — `x * gate(x)` (forward, line 24).
- **Intent:** **down-weight** the useless frequency dimensions and **up-weight** the good spatial ones, *adaptively per input* — a learned branch weighting. Gate weights come from the **whole 512-d vector**, so both domains condition the reweighting.



Illustrative SE-gate output: **spatial dims kept ≈ 0.9** , **frequency dims suppressed $\approx 0.2-0.3$** — but not to zero, which is why noise leaks through.

■ THE HONEST PUNCHLINE — SAY IT

In our results the SE gate **fails to fully suppress** the near-random frequency branch, so the hybrid still underperforms pure spatial. *Adding a gate was the right idea; it wasn't enough.* (Thesis §3.4.3 shows the gate suppressing low-value dims to $\sim 0.2-0.3$ and keeping high-value dims at ~ 0.9 .)

A classic trap: "BAB II says SE beats simple concatenation — but you use concatenation. Contradiction?" **No.**

- Order is **projection → CONCATENATION (512) → SE GATE → classifier**. Concatenation **unites** the two branches into one vector; SE gating **reweights** that vector afterward. They stack.
- "**Simple** concatenation" (the thing BAB II calls weaker) = concat with **no** adaptive weighting, straight into the classifier. Your model is on the **better** side of that comparison: concat **plus** SE gate.
- Analogy: "a plain wall is weak, so we reinforce it with steel." The wall (concat) stays; the steel (SE) is added. Not wall-vs-steel — **wall+steel**.

5

Hybrid training mechanics — the 3-group LR and freeze

scripts/train.py

1 HR

CORE

The hybrid is the trickiest to train — it mixes a **pretrained** backbone with a **randomly-initialized** freq branch and fusion head. Every choice protects the backbone.

CHOICE	VALUE	WHY (REHEARSE)
Backbone freeze	frozen first 3 epochs , unfreeze at epoch 4	Let FreqCNN + projections + SE + classifier stabilize before the pretrained backbone moves. Random parts backpropping into the backbone on epoch 1 would corrupt ImageNet features.
3-group differential LR	backbone $2e-5$ · freq $5e-5$ · head $2e-4$	Backbone precious → nudge gently. Freq is random → must move, not dominate. Head is random + on top → learns fastest. One high LR let random branches overwrite the backbone — a documented cause of the original collapse.
Optimizer	AdamW [45]	decoupled weight decay → cleaner regularization.
LR schedule	warmup 3 ep → cosine to $1e-6$	warmup avoids destabilizing the backbone on step 1; cosine anneals smoothly.
Loss	<code>BCEWithLogitsLoss</code> + <code>pos_weight</code> + label smoothing 0.05	<code>pos_weight</code> handles frame-level imbalance; smoothing curbs overconfidence on small data.
Consistent hflip	one flip → <code>TF.hflip(rgb)</code> and <code>torch.flip(fft)</code>	independent flips would desync the RGB and FFT views of the same frame.
Grad accumulation	2 steps · grad clip max-norm 5.0	stability on small effective batches.
Selection	best by val AUC , patience 12, max 30 ep	AUC = threshold-independent, robust selection.

IF ASKED “WHY THREE LEARNING RATES?”

“Three parameter groups with three roles: a pretrained backbone I must preserve (lowest LR), a randomly-initialized frequency branch that must

learn but not overpower the backbone (mid LR), and a randomly-initialized fusion head that should converge fast (highest LR). One flat LR let the random branches' early gradients wreck the ImageNet features — a real failure we fixed."

In-dataset AUC (mean over seeds 0/1/2)

	N=250	N=500	N=750
FFPP hybrid	0.540	0.616	0.644
CDF hybrid	0.787	0.839	0.919
FFPP spatial (baseline)	0.743	0.693	0.778
CDF spatial (baseline)	0.914	0.945	0.971

Ranking is **Spatial > Hybrid > Freq** on both datasets, every reliable tier. The hybrid **loses to its own baseline** — that IS the thesis.

Cross-dataset, n=750 (AUC / recall)

DIRECTION	MODEL	AUC	RECALL
FFPP→CDF	hybrid	0.665	0.599
FFPP→CDF	spatial	0.678	0.637
CDF→FFPP	hybrid	0.555	0.142
CDF→FFPP	spatial	0.607	0.074

Generalization drop ($\Delta F1$, n=750) — the ONE place hybrid helps

MODEL	TRAIN	F1 IN	F1 CROSS	Δ
spatial	FFPP	0.705	0.614	+0.091
hybrid	FFPP	0.606	0.594	+0.012 smallest
spatial	CDF	0.906	0.137	+0.769
hybrid	CDF	0.834	0.238	+0.597

● THE RM2 NUANCE — MEMORIZE THE EXACT SHAPE

FFT *does* reduce the generalization drop — **but only FFPP→CDF** ($\Delta F1$ +0.012 vs spatial's +0.091), and it's **bought with a lower in-dataset score** (0.606 vs 0.705). It does **not** hold CDF→FFPP. So FFT helps **inconsistently and at a cost** — the honest answer to RM2.

- The freq branch is $\approx$ **random** (AUC ~ 0.56 , flat val curve from epoch 1). Concatenating it **injects noise** into the fused 512-d vector.
- The **SE gate was meant to suppress** that noise but **can't fully** — so fusion $\leq$ spatial. The thesis (§4.2.1, §4.2.3): the frequency branch “*cenderung menjadi sumber noise yang menyeret turun performa fusi.*”
- **H0 cannot be rejected:** no meaningful cross-dataset generalization gain from FFT (BAB IV §4.2.1, BAB V §5.1).

■ CONTEXT — THE COLLAPSE DOC IS OLDER THAN THE FINAL MODEL

Hybrid_Model_Collapse_Analysis_2026-03-14.md describes the **pre-fix** hybrid: raw 2048:64 concat ($\sim 97\%$ spatial), per-image FFT min-max norm, no scheduler, no freeze, no differential LR $\rightarrow$ catastrophic collapse (CDF $\rightarrow$ FFPP F1 = 0.038, AUC 0.506). Your fixes — symmetric 256-d projections, SE gate, 3-group LR, freeze-then-unfreeze, cosine schedule, **global FFT z-score**, Youden's-J thresholds — **stopped the catastrophic collapse but did not make frequency informative**. That is the final, honest negative result.

IF ASKED “DID YOUR FIXES ACCOMPLISH ANYTHING?”

“Yes — they turned a catastrophic collapse (F1 0.038) into a competitive-but-not-better model. They did **not** make frequency discriminative. That localizes the limit to the **domain/representation**, not the training code — which *strengthens* the conclusion.”

Ranked by likelihood. HQ1 is the hardest question in the whole room, and it's aimed at you. **Recall each answer before revealing it.**

[REVEAL ALL](#)

HQ1 "Your proposed hybrid is worse than your own baseline. What's the contribution?"

+

CERTAIN · HARDEST IN THE ROOM

HQ2 "Explain your fusion. Why projection + concatenation + SE gating?"

+

ALMOST CERTAIN

HQ3 "What does the SE gate actually do here, and did it work?"

+

CERTAIN

HQ4 "BAB II says SE gating beats simple concatenation, but you use concatenation. Isn't that a contradiction?"

+

LIKELY TRAP

HQ5 "Is FFT a 4th channel of XceptionNet?"

+

LIKELY · EARLY-FUSION CONFUSION

HQ6 "Why does hybrid help FFPP→CDF but not CDF→FFPP?"

+

LIKELY

HQ7 "How do you know the hybrid isn't just under-trained or badly fused?"

+

LIKELY

HQ8 "Why not weight the branches manually, or drop the freq branch when it's bad?"

+

POSSIBLE

HQ9 "~25.9M parameters for a model that's worse than a 20.8M one — isn't that wasteful?"

+

POSSIBLE

HQ10 "Why late fusion and not early fusion?"

+

POSSIBLE

§3 Hybrid numbers card

Keep this on your phone. Every value from the results run.

HybridTwoBranch · Late Fusion + SE

RESULTS RUN

ARCHITECTURE

Xception(2048) + FreqCNN(512) → project both to 256 → concat 512 → **SE gate** → classifier

SE GATE

squeeze `Linear(512→128)` (reduction 4) → excite
`Linear(128→512)→Sigmoid` → `x*gate(x)`

CLASSIFIER

`Dropout(0.5) → Linear(512,128) → ReLU → Dropout(0.5) → L`

PARAMS

~25.9M (25,878,570) = spatial 20.8M + freq 4.2M + fusion ~0.85M

3-GROUP LR

backbone $2e-5$ · freq $5e-5$ (base $\times 0.25$) · head $2e-4$ · AdamW · warmup 3 → cosine to $1e-6$

FREEZE	spatial backbone frozen 3 ep → unfreeze epoch 4
IN-DATASET AUC	n=750 — FFPP 0.644 · CDF 0.919 — below spatial (0.778 / 0.971) both
CROSS-DATASET	n=750 — FFPP→CDF AUC 0.665 (recall 0.599) · CDF→FFPP AUC 0.555 (recall 0.142)
WHERE FFT HELPS	$\Delta F1$ FFPP→CDF +0.012 (smallest drop) vs spatial +0.091 — only this direction, at lower in-dataset F1
THRESHOLD (DEMO)	0.229 (Youden's J on FFPP val)
MECHANISM	freq $\approx$ noise → concat injects noise → SE gate can't fully suppress → hybrid $\leq$ spatial; H0 not rejected
KEY CITATIONS	[38] SE-Net/Hu · [6] Xception/Chollet · [5] He · [13] SpecXNet/Alam · [16] FSBI/Hasanaath · [45] AdamW · [46] Ben-David

§4 If you only have one evening

1 Draw the full forward pass with dims (2048 & 512 → 256 & 256 → 512 → SE → head) — HQ2.

2 Explain the SE gate (squeeze/excite/scale) and say it partially-but-not-fully suppresses the noise — HQ3.

3 Memorize the n=750 table (hybrid < spatial both) and the $\Delta F1$ +0.012 FFPP→CDF nuance — HQ1, HQ6.

4 Rehearse the 3-group LR + freeze rationale — HQ7, and the “concat + SE are both used” rebuttal — HQ4.

5 Own the negative: the fusion is well-built; the freq *input* is weak; hybrid $\leq$ spatial is the measured result. Re-read

`sidang_study_notes_detailed_2026-06-25.md` Steps 04, 05, 07.

§5 Literature notes worth keeping

■ BEFORE SIDANG

In-text numbers follow the docx. **Verify each [N] against Daftar Pustaka** — the fact-check flagged in-text / bibliography desync.

Hu et al. [38] — Squeeze-and-Excitation Networks

The SE gate's source. In BAB II §2.8 it's framed as an **adaptive multi-domain fusion mechanism** — channel attention that suppresses uninformative channels and enhances discriminative ones. Cite for the gate *and* the “better than simple concatenation” claim (which endorses your concat+SE design).

Chollet [6] (Xception) · He [5] (ResNet)

The two backbones inside the hybrid — depthwise-separable spatial branch, residual `FreqBlock`.

SpecXNet (Alam [13]) · FSBI (Hasanaath [16])

Dual-domain / freq-enhanced related work — cite as **inspiration**, and be precise: SpecXNet fuses spatial-spectral via channel split + Fourier attention inside each block; FSBI uses frequency-enhanced self-blended images. **Neither appends FFT as a 4th channel** — don't equate their approach with early fusion.

Ben-David et al. [46] — domain adaptation theory

The formal name for the cross-dataset drop the hybrid is trying (and mostly failing) to reduce.

AdamW (Loshchilov & Hutter) [45]

The optimizer for all three models.

Why the fused input is weak

For the mechanism behind $\text{hybrid} \leq \text{spatial}$, lean on the frequency guide's citations: Durall [8], Mejri [31], Oppenheim&Lim [25] / SPSSL [26], Geirhos [28] / Rahaman [29] / Wang [30].

§6 Known quirks to OWN before they're pointed out

■ "EARLY FUSION" LEFTOVERS IN THE .DOCX

The live Word doc still says "dan early fusion" in a couple of methodology spots and once frames FFT as a "4th channel." Those are **being revised** — the evaluated model is **late fusion only** (three models: spatial, freq, hybrid). If a reviewer quotes one: "that's the unevaluated early-fusion variant; our results are late fusion." (Fact-check Items 7 & 11.)

■ BATCHNORM IN THE PROJECTIONS NEEDS BATCH > 1

`spatial_proj` / `freq_proj` use `BatchNorm1d`; at inference the demo batches all frames together, so it's fine — just know it's there.

■ FREQ CLASSIFIER BYPASSED

Inside the hybrid, only `FreqCNN.features` (512-d) is used; the standalone freq classifier head is not part of the hybrid.

■ "SE LEARNS CORRELATIONS" – DON'T OVERCLAIM

SE gating is **channel reweighting conditioned on both domains**, not full pairwise cross-attention / bilinear correlation. Say "adaptive channel weighting conditioned on the fused vector," not "models the full spatial-frequency correlation."

■ EFFECTIVE BATCH VARIES BY GPU

The canonical run auto-tunes batch size; BAB IV's "batch efektif 32" is the config baseline. If asked: "effective batch was GPU-dependent via auto-tuning; the config baseline is 32."

§7

The demo — hybrid-specific things a reviewer can point at live

Pair with `sidang_study_guide_DEMO_debug.md`. This is **only** the hybrid angles. The demo runs all three models side-by-side so the comparison is self-evident — your model is the *proposal*, and it's supposed to sit at or below spatial.

What is *yours* on screen

- The **hybrid verdict card** — fake-probability bar with a tick at threshold **0.229**. In `inference.predict_video` the hybrid path is `model(rgb_batch, fft_batch)` (consumes **both** inputs), then `sigmoid(logits).mean()` over frames.
- The comparison itself is the point: **hybrid tracking close to but not above spatial is the thesis made visible** — not a bug.

Hybrid-specific failure modes

SYMPTOM	CAUSE	WHAT TO SAY
Hybrid agrees-but-underperforms or loses to spatial	the finding — freq input is near-noise, SE can't fully suppress it	"Expected and central to proposed hybrid does not baseline. A measured co
Card shows "⚠ FFT calibration estimated"	<code>fft_stats.json</code> missing → fallback mean 5.0/std 3.0 (<code>inference.py:123-133</code>)	Hybrid depends on FFT branch). Copy <code>outputs/fft_cache/FFP</code> → <code>checkpoints/</code> . Spati
<code>RuntimeError: size mismatch loading hybrid.pt</code>	built arch ≠ checkpoint (wrong <code>freq_depth</code> / <code>base_channels</code>)	The demo reads depth/b checkpoint's embedded (<code>inference.py:65-66</code> , c Copy the right run's <code>bes</code>
Hybrid disagrees with both on an OOD clip	out-of-distribution + a noisy freq branch	"Models are trained on F sources are OOD, and th noise — that's the fusion

Reviewer: "change X and show me" — hybrid edit points

- "Run just the hybrid" → keep only the `("hybrid", ...)` row in `inference.py` `MODELS_SPEC` .
- "Use 0.5 threshold instead of 0.229" → delete/rename `hybrid_threshold.json` ; it falls back to 0.5 (`inference.py:118-119`). Good moment to explain Youden's J.
- Anything touching FFT preprocessing (fps, crop, high-pass) affects the hybrid's freq branch → ▲ IT WILL NO LONGER MATCH THE TRAINING DISTRIBUTION, SO THE NUMBERS STOP BEING COMPARABLE TO THE THESIS.

● THE ONE SENTENCE FOR THE DEMO

"Watch the hybrid sit right at or just under the spatial card on every clip — that side-by-side *is* our result: a well-engineered fusion can't turn a near-random frequency input into an improvement, so the honest answer to 'does FFT help?' is no under these conditions."

COMPANION GUIDES

SPATIAL branch

FREQUENCY branch

◆ HYBRID branch

DEMO debug

Sidang study guide · HybridTwoBranch late fusion · deepfake_hybrid · values from the
results run